

Session 1.2: Network Effects & Penguin Effect

Direct/indirect network externalities, adoption coordination dilemmas, and Blu-ray vs. HD-DVD.

PART 1 (LECTURE)
60 Minutes

PART 2 (LAB PRACTICE)
60 Minutes

PART 3 (EXECUTIVE MEMO)
60 Minutes

STANDARD
Audited SEC Filings

Learning Objectives & Executive Overview

Clear conceptual grounding for undergraduate and graduate business students.

CORE COMPETENCIES

Goals

What You Will Master Today

- **Theoretical Mastery:** Evaluate seminal literature and analytical frameworks governing platform competition.
- **Empirical Rigor:** Reconcile management strategic claims with legally audited statutory filings.
- **Data Modeling:** Execute spreadsheet calculations with explicit coordinates and formula logic.
- **Executive Formulation:** Deliver structured board-level recommendations resolving complex trade-offs.

NETWORK EXTERNALITIES

Vocabulary

Key Topical Foundations

- **Metcalfé's Law::** Network value scales with user base squared (N^2).
- **Katz & Shapiro::** Adoption utility increases with total network size.
- **Penguin Effect::** Farrell & Saloner on adoption coordination freeze.
- **Micro-Examples::** Messaging apps, gaming consoles, office software.

Theoretical Foundations: Network Externalities

Rigorous analytical frameworks and literature benchmarks.

THEORETICAL ARCHITECTURE

Literature

Seminal Framework Insights

- *Metcalfe's Law*:: Network value scales with user base squared (N^2).
- *Katz & Shapiro*:: Adoption utility increases with total network size.
- *Penguin Effect*:: Farrell & Saloner on adoption coordination freeze.
- *Micro-Examples*:: Messaging apps, gaming consoles, office software.

STRATEGIC APPLICATION

Analysis

Business Model Mechanics

- *Format War (2006-08)*:: Sony Blu-ray vs. Toshiba HD-DVD standoff.
- *Coordination Trap*:: Consumers paused hardware purchases.
- *Trojan Horse Strategy*:: Bundling Blu-ray drives into Sony PS3.
- *Tipping Point*:: Movie studios committed exclusively to Sony.

STRATEGIC TAKEAWAY

Direct/indirect network externalities, adoption coordination dilemmas, and Blu-ray vs. HD-DVD.

Enterprise Case Dissection: Blu-ray vs. HD-DVD

Historical case evidence and multi-year corporate trajectory analysis.

CORPORATE DYNAMICS

Timeline

Market Conditions & Strategic Moves

- **Format War (2006-08)::** Sony Blu-ray vs. Toshiba HD-DVD standoff.
- **Coordination Trap::** Consumers paused hardware purchases.
- **Trojan Horse Strategy::** Bundling Blu-ray drives into Sony PS3.
- **Tipping Point::** Movie studios committed exclusively to Sony.

ANALYTICAL FOCUS

Evidence

Operational & Financial Findings

- **Sony 20-F Data::** Game segment sales & operating income (FY06-10).
- **Operating Margin::** Margin % calculation ($=D2/C2*100$).
- **Hardware Scale::** Plotting cumulative hardware vs. profitability.
- **Finding::** Tipping achieved at 50M units; FY10 turned +35.6B JPY.

Lab Dataset & Empirical Metrics: Applied Excel Lab

Reconciling corporate filings with explicit spreadsheet formulas.

SPREADSHEET MODELING

Lab Spec

Data Metrics & Formulas

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MANAGEMENT DILEMMA

Decision Lab

Strategic Tension

- **Subsidization Risk::** Cumulative hardware losses exceeded 400B JPY.
- **Balance Sheet Limit::** Startup survival under heavy cash burn.
- **Alternative Paths::** Single-user utility & exclusive launch anchors.
- **Strategic Memo::** Defending hardware subsidies to shareholders.

Executive Problem Solving: Executive Lab

Framing complex management dilemmas and evaluating corporate trade-offs.

EXECUTIVE CHALLENGE

Dilemma

Corporate Strategic Tension

- **Subsidization Risk::** Cumulative hardware losses exceeded 400B JPY.
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EVALUATION RUBRIC

Requirements

Board-Level Deliverable

- **Actionable Recommendation:** Formulate a concrete 3-bullet decision brief for senior leadership.
- **Quantitative Defense:** Support recommendations using audited financial metrics from the lab spreadsheet.
- **Risk Mitigation:** Address channel friction, partner resistance, or competitive retaliation.
- **Implementation Timeline:** Define immediate 90-day execution milestones.

Lab Transition: 120-Minute Hands-On Execution

Transitioning from lecture observation to hands-on data manipulation and strategic decision-making.

EXERCISE 1: FOUNDATIONAL PRACTICE

60 Min Lab

Applied Excel Lab

Hands-on spreadsheet protocol using audited historical numbers:

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EXERCISE 2+: ADVANCED PROBLEM SOLVING

60 Min Memo

Executive Lab

Applied scenario analysis, trade-off evaluation, and executive brief:

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